

Sustainable Smart City Assistant Using IBM Granite LLM

1. INTRODUCTION

1.1. Project Overview

A sustainable smart city assistant is an intelligent system that leverages technology to improve the quality of life and resource management while promoting environmental sustainability. It encompasses various applications like AI-powered energy management, personalized citizen services, and optimization of city operations. Essentially, it's about using technology to make cities more efficient, liveable, and environmentally friendly.

1.2. Purpose

The purpose of a Sustainable Smart City Assistant is to empower cities and their residents to thrive in a more eco-conscious and connected urban environment. By leveraging AI and real-time data, the assistant helps optimize essential resources like energy, water, and waste, while also guiding sustainable behaviours among citizens through personalized tips and services. For city officials, it serves as a decision-making partner—offering clear insights, forecasting tools, and summarizations of complex policies to support strategic planning. Ultimately, these assistant bridges technology, governance, and community engagement to foster greener cities that are more efficient, inclusive, and resilient.

2. IDEATION PHASE

2.1. Problem Statement

Urban residents and city administrators struggle to access real-time, actionable insights about sustainability metrics such as energy usage, pollution levels, and policy updates. Existing systems are fragmented, non-intuitive, and lack intelligent support for decision-making or citizen engagement. There is a need for a unified, AI-powered assistant that can simplify complex data, provide personalized eco-advice, and foster transparent communication between citizens and city services.

2.2. Empathy Map Canvas

An empathy map is a simple, easy-to-digest visual that captures knowledge about a

user's behaviours and attitudes. It is a useful tool to help teams better understand their users. Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

2.3. Brainstorming

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich number of creative solutions.

3. REQUIREMENT ANALYSIS

3.1. Customer journey map

Creating a customer journey map for a sustainable smart city assistant involves visualizing how citizens interact with the assistant across various stages-while embedding sustainability at every touchpoint. Here's a simplified outline to get you started

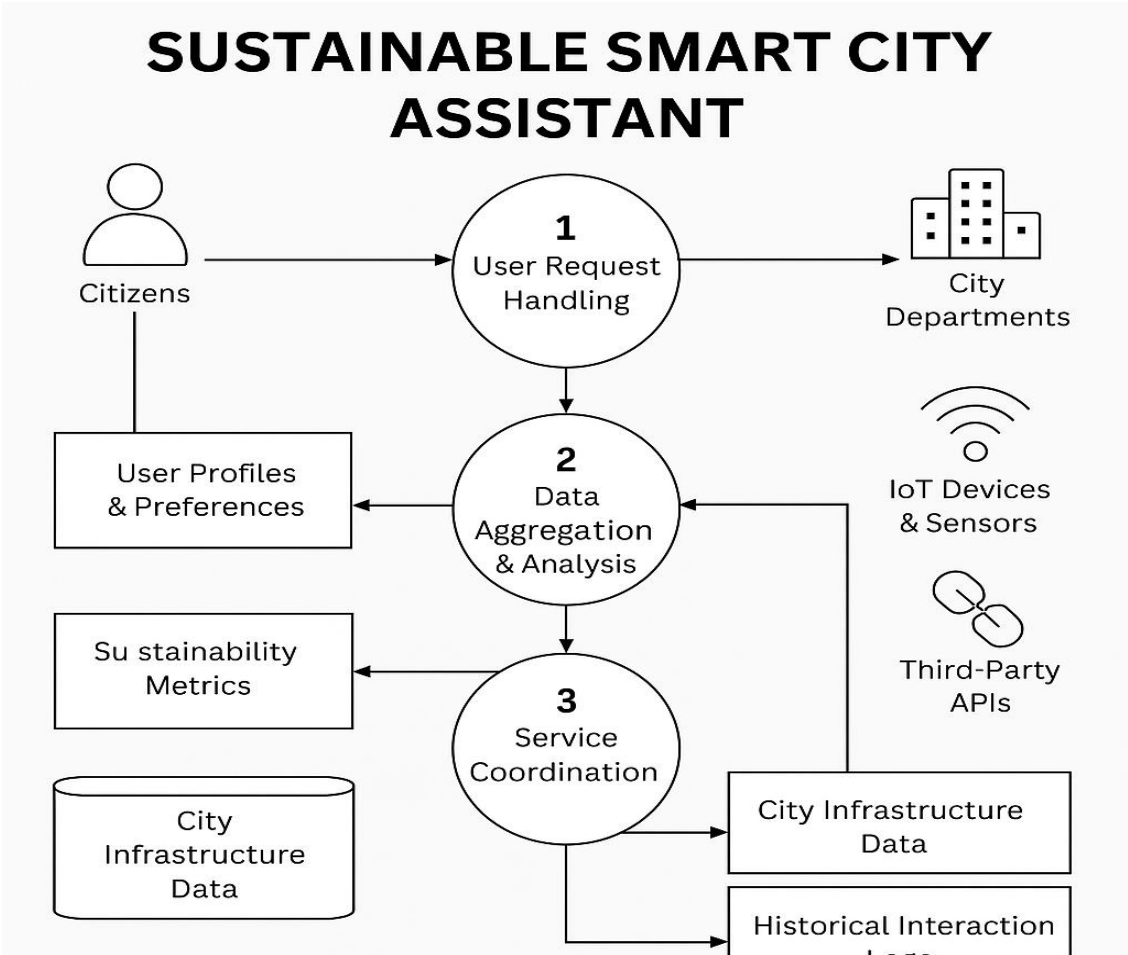
CUSTOMER JOURNEY MAP: SUSTAINABLE SMART CITY ASSISTANT			
Stage	Citizen Goal	Touchpoints	Sustainability Focus
 Awareness	Discover the assistant	Social media, city website, community events	Promote eco-benefits (e.g. reduced energy use, green tips)
 Onboarding	Learn how it helps	App store, demo videos, peer reviews	Highlight transparency, data privacy and green features
 Engagement	Use for daily tasks (e.g., transport, waste mgmt)	Voice/text chat, push notifications, dashboards	Encourage eco-friendly choices (e.g, public transport recycling)
 Advocacy	Resolve issues or ask questions	Help center, live chat, feedback forms	Promote digital-first support to reduce resource use
	Recommend to others	Social sharing, community	Reward sustainable

3.2. Solution Requirement

The proposed solution for the Sustainable Smart City Assistant involves building an AI-powered platform using IBM Granite LLM to enhance urban sustainability and citizen engagement. It integrates real-time data from IoT sensors with natural language capabilities to provide services like policy summarization, eco-tip generation, forecasting, and anomaly detection.

3.3. Data Flow Diagram

A Sustainable Smart City Assistant is a digital tool designed to help citizens and city officials make eco-conscious decisions using real-time data and intelligent systems. It integrates technologies like AI, IoT, and cloud computing to manage urban services such as transport, energy, waste, and water—more efficiently and sustainably.



3.4. Technology stack

To build a Sustainable Smart City Assistant, you need a robust and scalable technology stack that integrates real-time data, intelligent decision-making, and

citizen engagement.

Layer	Technologies / Tools	Purpose
User Interface (UI)	Streamlit, React, Flutter	Interactive dashboards, mobile/web apps for citizen interaction.
Application Layer	FastAPI, Node.js, Django	Business logic, API routing, service orchestration.
AI & Analytics	IBM Watsonx, Scikit-learn, TensorFlow, PyTorch	Natural language processing, forecasting, anomaly detection.
Data Layer	PostgreSQL, MongoDB, Pinecone (vector DB), FPDF/Markdown	Structured/unstructured data storage, document generation.
IoT & Sensors	Arduino, Raspberry Pi, LoRaWAN, Zigbee, MQTT	Real-time data collection from city infrastructure.
Cloud & Edge	AWS, Azure, Google Cloud, EdgeX Foundry	Scalable computing, edge processing for low-latency responses.
Security & Privacy	OAuth 2.0, SSL/TLS, Blockchain, GDPR-compliant frameworks	Data protection, secure transactions, citizen trust.
Integration Layer	RESTful APIs, WebSockets, GraphQL	Seamless communication between services and third-party systems.
Monitoring & DevOps	Prometheus, Grafana, Docker, Kubernetes, Jenkins	System health, containerization, CI/CD pipelines.

This stack enables the assistant to deliver real-time insights, support sustainable urban services, and ensure secure, inclusive citizen engagement.

4. PROJECT DESIGN

4.1. Problem Solution Fit

- Citizens lack real-time access to eco-friendly transport and energy data
→ The assistant provides live updates on green transit, energy usage, and sustainability tips.
- City departments face fragmented data and delayed responses
→ The assistant integrates IoT data and automates alerts and service coordination.

- Residents are unaware of local sustainability programs or incentives
→ The assistant sends personalized notifications about green initiatives and rewards.
- Reporting urban issues (like waste overflow) is difficult and time-consuming
→ The assistant enables voice/text-based issue reporting with automated routing to departments.
- There's no feedback loop for continuous improvement
→ The assistant collects user feedback and usage data to refine and evolve services.

4.2. Proposed Solution

The proposed solution for the Sustainable Smart City Assistant involves building an AI-powered platform using IBM Granite LLM to enhance urban sustainability and citizen engagement. It integrates real-time data from IoT sensors with natural language capabilities to provide services like policy summarization, eco-tip generation, forecasting, and anomaly detection.

- **AI-Powered Citizen Assistant**

A multilingual chatbot and voice assistant that helps citizens access services, report issues, and receive sustainability tips in real time.

- **IoT-Driven Urban Monitoring**

Integration with smart sensors for air quality, traffic, energy, and waste to provide real-time data for decision-making.

- **Personalized Sustainability Dashboard**

A mobile/web interface showing individual and community-level metrics like energy use, water consumption, and carbon savings.

- **Green Mobility Integration**

Real-time updates on public transport, EV charging stations, and bike-sharing options to promote low-emission travel.

- **Smart Waste & Water Management**

Automated alerts for waste overflow, leak detection, and optimized collection routes using predictive analytics.

- **Community Engagement & Rewards**

Gamified features that reward eco-friendly behaviour (e.g., recycling, using public transport) with points or local incentives.

- **Open Data & Transparency**

Public dashboards and APIs to foster trust, innovation, and third-party solution development.

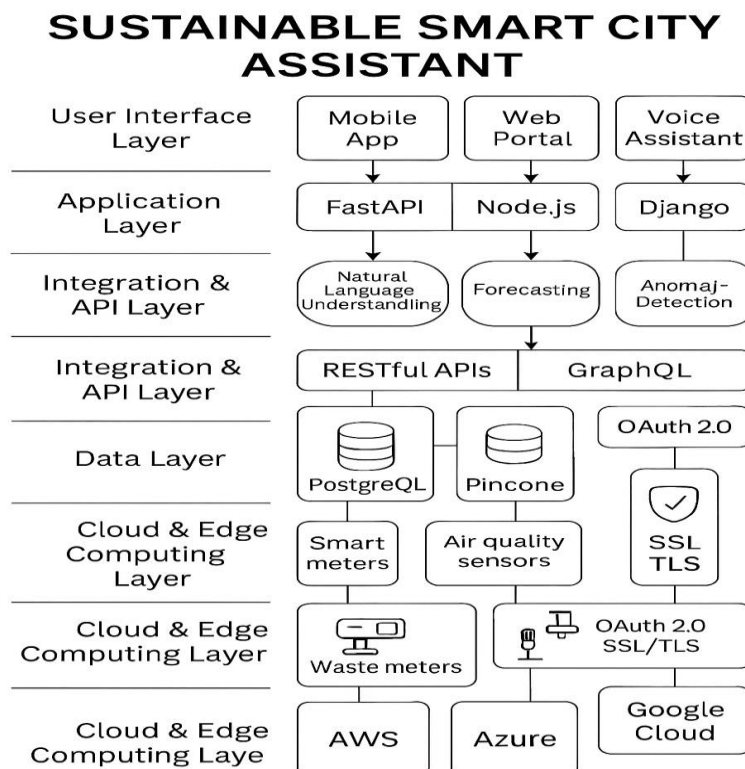
- **Security & Privacy by Design**

End-to-end encryption, GDPR-compliant data handling, and user-controlled data sharing preference

4.3. Solution Architecture

The solution architecture for the Sustainable Smart City Assistant using IBM Granite LLM includes a user-friendly interface (Streamlit/Gradio), a FastAPI backend connected to Granite for chat and summarization, real-time IoT data integration, a vector database for semantic search, and secure cloud deployment with monitoring tools. It's modular, scalable, and designed for responsive, AI-driven urban services.

Technical Architecture:



- **User Interaction Layer**

Interfaces: Mobile app, web portal, voice assistant

Purpose: Enables citizens to access services, report issues, and receive sustainability tips

- **Application & Service Layer**

Technologies: FastAPI, Node.js, Django

Functions: Handles business logic, workflows, and service orchestration

- **AI & Intelligence Layer**

Tools: IBM Watsonx, TensorFlow, Scikit-learn

Capabilities: Natural language understanding, forecasting, anomaly detection, recommendations

- **Data Management Layer**

Databases: PostgreSQL, MongoDB, Pinecone

Stores: User profiles, sensor data, sustainability metrics, feedback logs

- **IoT & Sensor Integration Layer**

Devices: Smart meters, air quality sensors, waste bin sensors

Protocols: MQTT, LoRaWAN, Zigbee

Role: Real-time data collection from city infrastructure

- **Integration & API Layer**

Interfaces: RESTful APIs, GraphQL, WebSockets

Purpose: Connects internal modules and third-party services (e.g., weather, maps)

- **Cloud & Edge Computing Layer**

Platforms: AWS, Azure, Google Cloud, EdgeX Foundry

Role: Scalable processing, low-latency responses, secure storage

- **Security & Governance Layer**

Technologies: OAuth 2.0, SSL/TLS, Blockchain (optional)

Policies: GDPR compliance, access control, audit logs

5. PROJECT PLANNING & SCHEDULING

5.1. Project Planning

Here's a completed Product Backlog, Sprint Schedule, and Estimation table for our Sustainable Smart City Assistant using IBM Granite LLM, continuing from your format and including the Dashboard epic:

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	2	High
Sprint-1		USN-2	As a user, I will receive confirmation email once I have registered for the application	1	High
Sprint-2		USN-3	As a user, I can register for the application through Facebook.	2	Low
Sprint-1		USN-4	As a user, I can register for the application through Gmail.	2	Medium
Sprint-1	Login	USN-5	As a user, I can log into the application by entering email & password.	1	High

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1. Functional Testing

- Verify user interactions (voice/text) return accurate and relevant responses.
- Test integration with external APIs (e.g., weather, transport, energy).
- Validate real-time data processing from IoT sensors (e.g., air quality, waste bins).
- Ensure workflows like issue reporting and service scheduling function end-to-end.
- Confirm authentication, data encryption, and privacy compliance (e.g., GDPR).

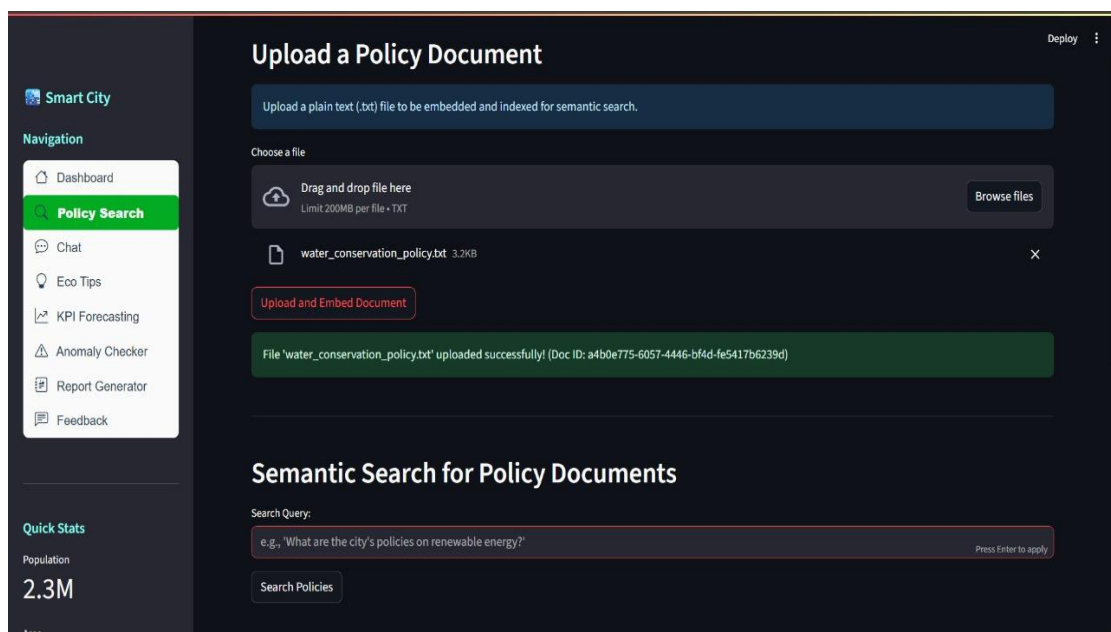
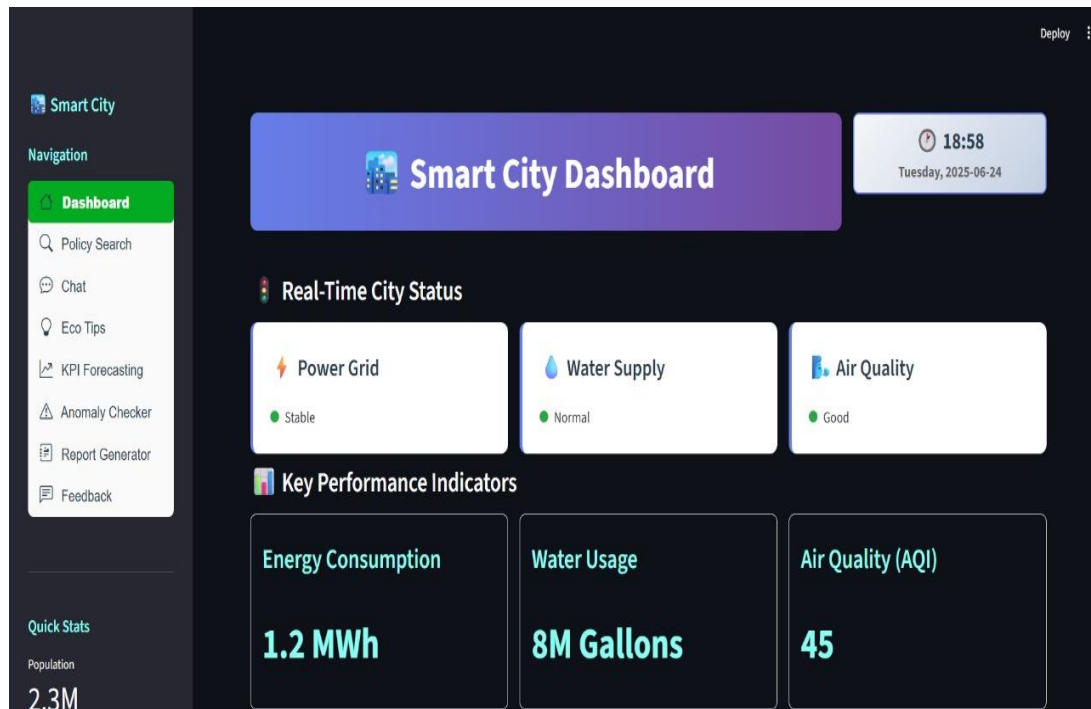
6.2. Performance Testing

- Conduct load testing to simulate high user traffic and ensure stability.
- Perform stress testing to evaluate system behavior under extreme conditions.
- Measure latency for real-time services like traffic or energy updates.

- Test scalability with increasing IoT data and user base.
- Monitor system resource usage (CPU, memory, bandwidth) under various loads.

7. RESULTS

7.1. Output Screenshots



Smart City

Navigation

Dashboard

Policy Search

Chat

Eco Tips

KPI Forecasting

Anomaly Checker

Report Generator

Feedback

Quick Stats

Population

2.3M

Area

Deploy

Chat with your Sustainable AI Assistant

🗨️ Hello! I'm your Sustainable City AI Assistant. 🌱 I'm here to support your journey toward building smarter, greener, and more resilient cities. Whether you need insights on energy consumption, water usage, waste management, or air quality, I'm ready to help you make data-driven decisions for a sustainable future.

🗨️ what is sustainable city?

🗨️ A sustainable city is an urban center designed with long-term environmental, economic, and social sustainability in mind. Key characteristics include efficient energy use, reduced carbon emissions, waste minimization, green infrastructure, mixed-use development, accessible public transportation, and promotion of social equity and well-being. Such cities typically integrate smart technologies for improved urban planning, governance, and resource management.

🗨️ 5 steps to maintain sustainalbe city

🗨️

- **Promote Renewable Energy**:** Encourage the use of renewable energy sources like solar, wind, and hydroelectric power for city infrastructure and buildings. This reduces dependence on fossil fuels and decreases greenhouse gas emissions.
- Improve Public Transportation:** Develop and enhance public transportation systems such as buses, trams, and trains, making them efficient, affordable, and accessible. This decreases private vehicle usage, thereby reducing air pollution and traffic congestion.
- Green Spaces and Urban Forestry:** Increase and maintain green spaces, parks, and urban forests in the city. These elements help in air purification, regulating temperature, absorbing carbon dioxide, and providing habitats for wildlife.

Ask me anything...

➔

Smart City

Navigation

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Policy Search

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2.3M

Area

Deploy

resources.

5. **Smart Building and Energy-Efficient Technologies:** Encourage the construction and renovation of buildings to be energy-efficient. Smart building technology, like automated lighting and HVAC systems, can optimize resource consumption based on occupancy and environmental conditions, further reducing a city's ecological footprint.

These steps, when consistently applied and adapted to a city's unique characteristics and resources, can help maintain a sustainable urban environment.

🗨️ what is covid 19?

🗨️ I'm here to help with questions about sustainable cities, smart city governance, and urban living. Please ask something related to these topics!

🗨️ suggest me some tips to maintain good health

🗨️ I'm here to help with questions about sustainable cities, smart city governance, and urban living. Please ask something related to these topics!

🗨️ give me some educational tips.

🗨️ I'm here to help with questions about sustainable cities, smart city governance, and urban living. Please ask something related to these topics!

Ask me anything...

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Smart City

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Get Personalized Eco-Friendly Tips

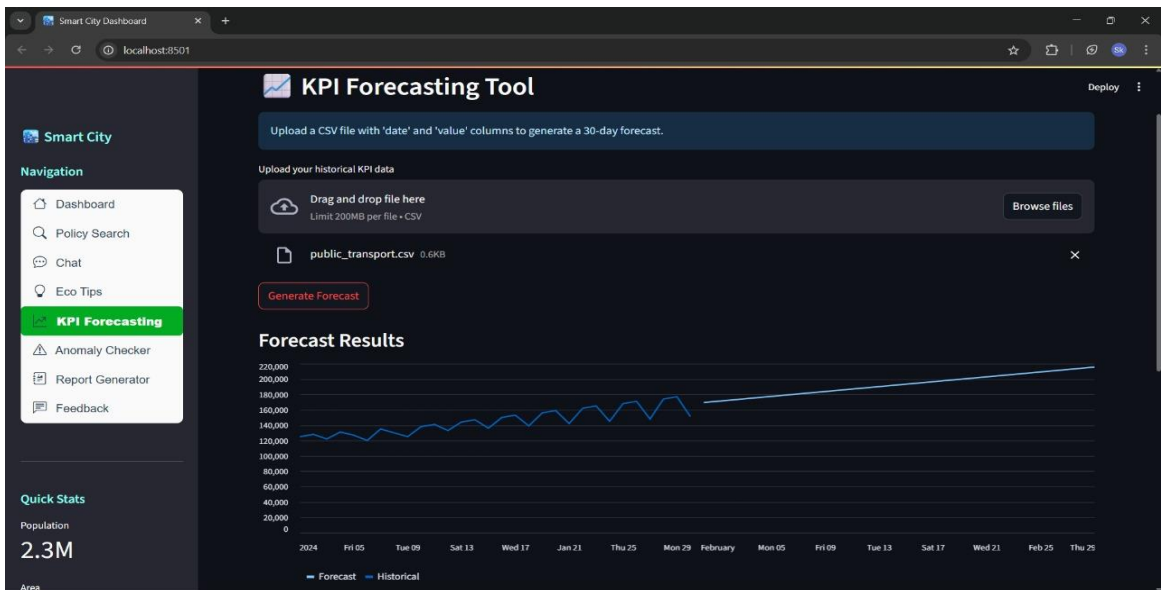
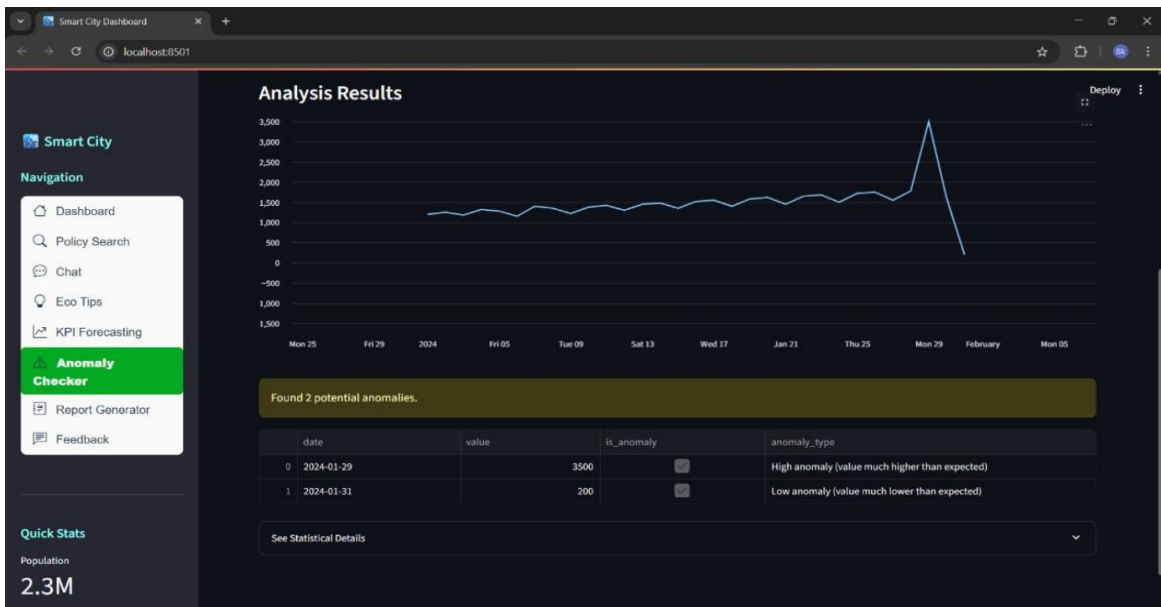
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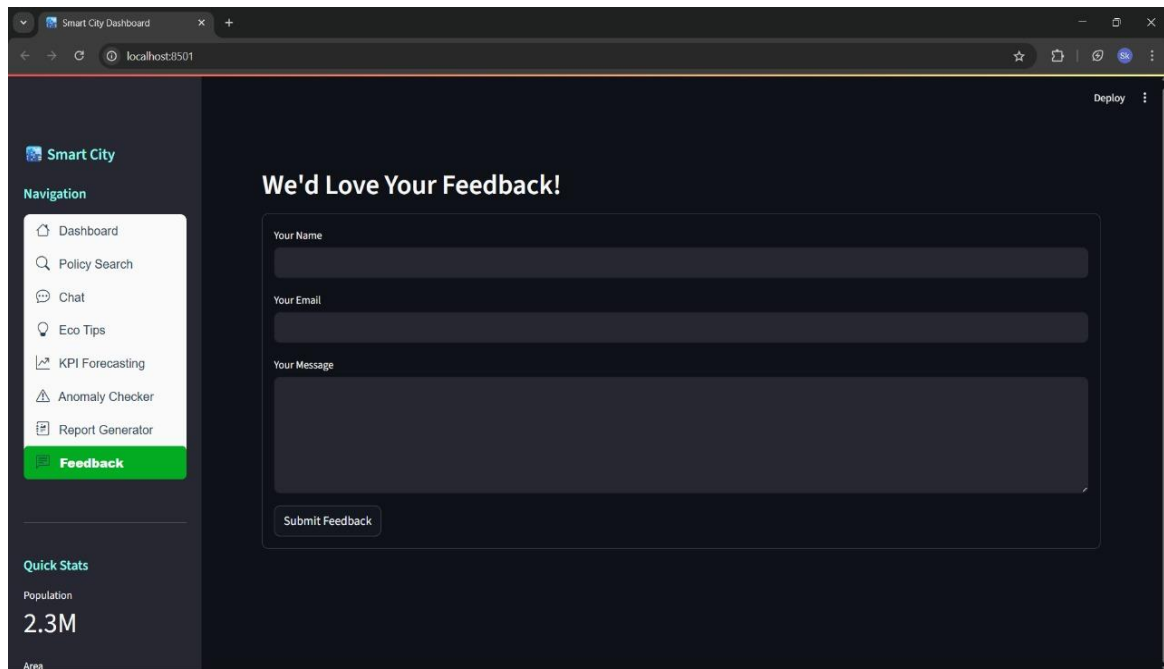
water

Get Eco Tip

🌱 Eco Tip for 'Water':

- Implement Rainwater Harvesting:** Encourage buildings, both residential and commercial, to install rainwater harvesting systems. This captured rainwater can be used for non-potable purposes like irrigation, toilet flushing, and even car washing. City-wide initiatives, such as offering subsidies or tax benefits for installing such systems, can significantly increase participation.
- Establish Decentralized Water Treatment and Reuse:** Deepen community involvement in local water management by setting up small-scale, decentralized wastewater treatment and reuse systems. These could be implemented in neighborhoods or districts, treating wastewater for non-potable uses like irrigation of public spaces, street cleaning, or even cooling towers for industrial processes. This not only reduces strain on centralized infrastructure but also fosters local sustainability consciousness and resilience.





8. ADVANTAGES & DISADVANTAGES

8.1. Advantages

1. Improves Environmental Awareness
2. Optimizes Resource Usage
3. Encourages Eco-Friendly Behavior
4. Supports Smarter Urban Planning
5. Enhances Quality of Life
6. Promotes Community Engagement
7. Increases Resilience to Climate Change
8. Reduces Operational Costs
9. Boosts Innovation and Green Economy
10. Empowers Decision-Ma

8.2. Disadvantages

1. High Implementation Costs
2. Privacy and Data Security Concerns
3. Digital Divide and Accessibility Issues
4. Dependence on Technology
5. Maintenance and System Failures

6. Resistance to Behavioral Change
7. Complex Integration with Existing Infrastructure
8. Risk of Data Misuse or Surveillance
9. Limited Effectiveness Without Public Participation
10. Need for Continuous Updates and Improvements

9. CONCLUSION

The Sustainable Smart City Assistant represents a transformative step toward greener, more efficient urban living. By integrating AI, IoT, and real-time data, it empowers citizens to make eco-conscious choices and enables city officials to optimize services with precision. This assistant not only bridges the gap between technology and sustainability but also fosters transparency, inclusivity, and resilience in urban ecosystems. As cities continue to grow, such intelligent systems will be essential in shaping a future that is both livable and sustainable. Its advantages such as optimizing resource use, raising environmental awareness, and enhancing quality of life make it a powerful tool for modern urban development. However, its effectiveness depends on thoughtful implementation. Challenges like high costs, data privacy concerns, and unequal access must be addressed to ensure inclusive and ethical use. Additionally, the success of such a system relies heavily on public participation and continuous technological updates.

10. FUTURE SCOPE

• AI-Driven Urban Planning

Predictive models and digital twins will simulate urban growth, optimize zoning, and reduce environmental impact.

• Hyper-Personalized Citizen Services

Assistants will adapt to individual behaviors, offering tailored sustainability tips, transport suggestions, and energy-saving nudges.

• Autonomous Infrastructure Management

Integration with robotics and AI will enable self-healing infrastructure—like automated road repairs or smart waste drones.

• Climate Resilience & Disaster Response

Real-time data and AI will help cities predict, prepare for, and respond to climate events like floods, heatwaves, or wildfires.

- **Green Economy Integration**

Assistants will connect citizens and businesses to carbon credit systems, green job opportunities, and circular economy platforms.

- **Ethical AI & Digital Inclusion**

Future systems will prioritize fairness, accessibility, and transparency—ensuring all citizens benefit, regardless of age or ability.

- **Cross-City Collaboration**

Smart assistants may form networks across cities, sharing anonymized data to benchmark sustainability and co-develop solution

11. APPENDIX

11.1. GitHub Project Link

<https://github.com/venu2024/Sustainable-Smart-City-Assistant.git>

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